

Analysis of Statistical Similarity of Network Services

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14. června 2023

- Dataset - HTTPS network flows from `ipfixprobe`
- 1 287 422 rows/flows, 12 attributes
- service name `TLS_SNI` + flow metadata
- 19 910 unique services

- remove (partially) unfinished flows - 177 377
- engineer new attributes from existing ones: flow duration (s), avg packet size (B), avg packet size rev (B), total packets, total bytes (B), time between packets (s), traffic speed (B/s), traffic ratio (upload/download)
- rename not so unique 'TLS_SNI', reducing unique values down to 7 881

```
÷ string TLS_SNI
10 acd460827b1ac2fa2b4d1fbdaaae3cfb.safeframe.googlesyndication.com
77 38cea87750dffeadd7edc9d62fb07ff6f.safeframe.googlesyndication.com
119 0ce29eea4ad84dfcbc27b6bb3c3d5dc2.safeframe.googlesyndication.com
576 21e6a1386772626f9f650b9bab7534fd.safeframe.googlesyndication.com
630 4a398fe09e62caf5e60a38c34e4915bf.safeframe.googlesyndication.com
```

- basic statistical data (`df.describe()`)
- services with the most flows

```
string TLS_SNI
www.google.com        62947
dns.google             29130
fonts.gstatic.com      27281
doh.opendns.com        26603
safebrowsing.googleapis.com 21391
www.youtube.com      21010
play.googleapis.com    20825
geller-pa.googleapis.com 20653
play.google.com        19500
googleads.g.doubleclick.net 19345
```

- some services with the least flows

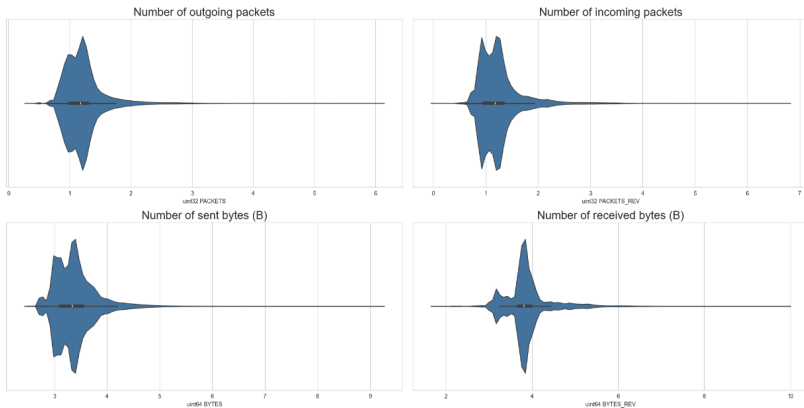
```
string TLS_SNI
www.mcourser.cz 1
ket.sk 1
aas.medialaben.no 1
waze.com 1
www.gli.cas.cz 1
www.rodinnebaleni.cz 1
www.bootyshop.eu 1
amp-xhamster-com.cdn.ampproject.org 1
www.zigwheels.my 1
odeholm-audio.com 1
```

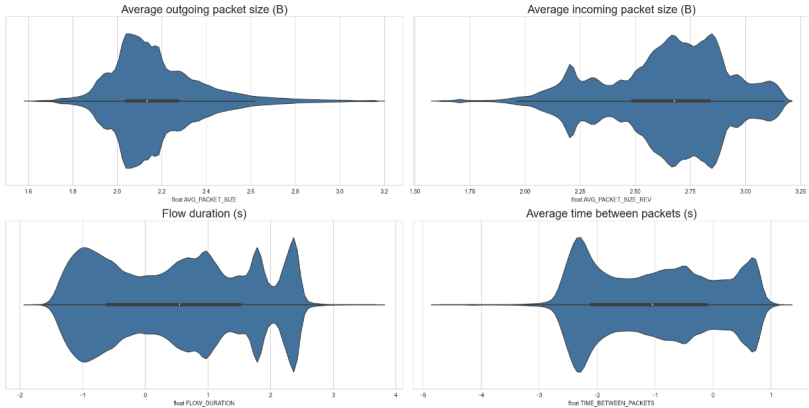
- services transferring the most data

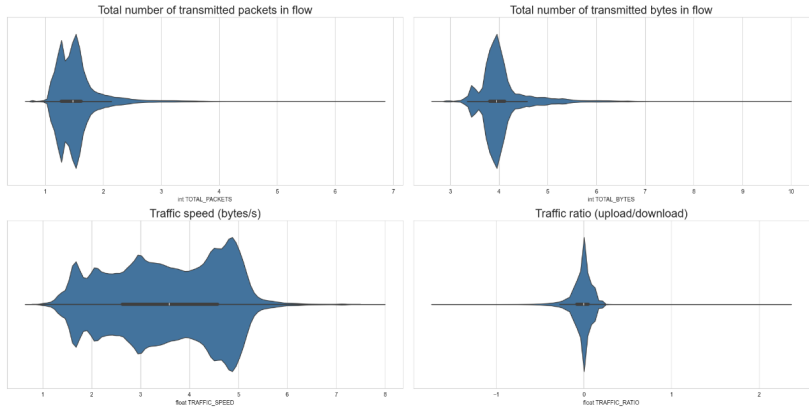
string TLS_SNI	int TOTAL_BYTES
googlevideo.com	17210344582
doc-04-0a-docs.googleusercontent.com	8718081652
www.googleapis.com	7423526669
www.jh-inst.cas.cz	6592265764
8.dl.wsfiles.cz	5653964611
cz-bks400-prod11-live.solocoo.tv	5216431656
c.2mdn.net	4617113604
www.google.com	3659352897
mail.google.com	3572539331
cz-bks400-prod15-live.solocoo.tv	3414582852

- services transferring the least data

string TLS_SNI	÷ int TOTAL_BYTES ÷
psychoklub.com	1086
sula.sk	1008
user-data-storage-stickman-hook.madbox.io	973
www.panickedravce.sk	957
www.scotthegerty.com	956
static-dmp.mediaglacier.com	941
elamir-steel.com	925
netpincer-1239.firebaseio.com	833
ecommerce.iap.unity3d.com	688
posthog.youversion.com	596

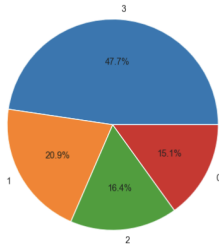






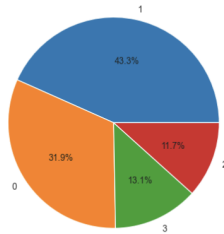
- preprocessing, feature engineering
- dropping potentially irrelevant attributes
- working with dataset 1 110 045 x 14
- PCA + k-Means - 8 PCA, 4 clusters

Cluster distribution for googlevideo.com



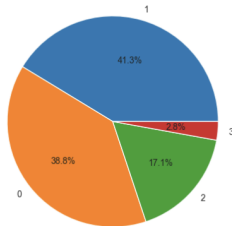
Total: 7240

Cluster distribution for www.youtube.com



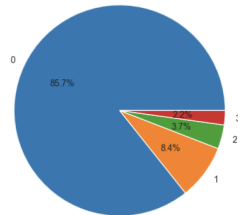
Total: 21010

Cluster distribution for dns.google



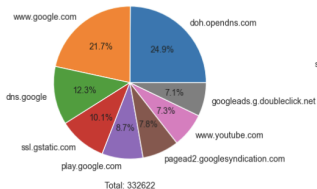
Total: 29130

Cluster distribution for doh.opendns.com

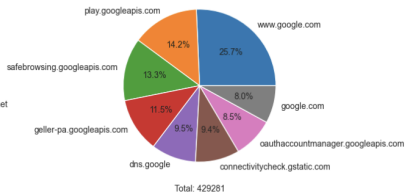


Total: 26603

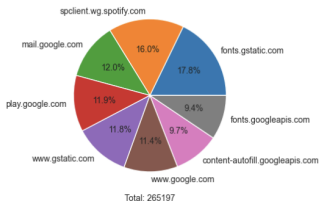
TLS_SNI distribution for Cluster 0



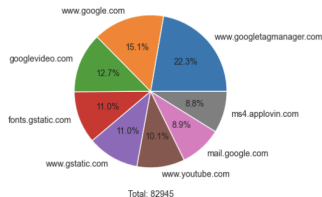
TLS_SNI distribution for Cluster 1



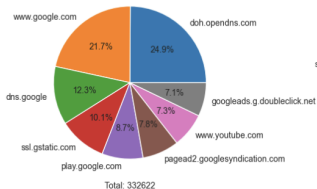
TLS_SNI distribution for Cluster 2



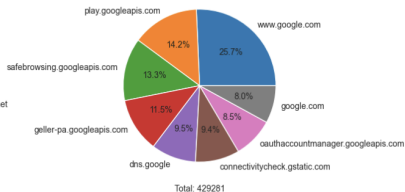
TLS_SNI distribution for Cluster 3



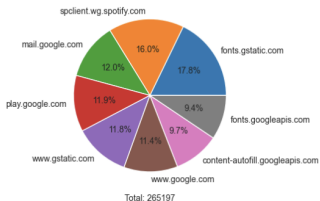
TLS_SNI distribution for Cluster 0



TLS_SNI distribution for Cluster 1



TLS_SNI distribution for Cluster 2



TLS_SNI distribution for Cluster 3

